

CE 310

Week 16 — Capstone Presentations

Cross-Team Comparison and Course Debrief

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Presentation Format

5 minutes per group — 4 min presentation + 1 min Q&A

Each presentation should cover four points:

1. **Dataset & client profile** — what's the data, who's the client, key descriptive stats
2. **Regression results** — Model 2 slope and GROUP coefficient, what they mean for your topic
3. **Hypothesis test** — is the group effect statistically significant? Cohen's d?
4. **Uncertainty** — Monte Carlo P10/P90, one actionable recommendation

Schedule and Debrief Timing

Every team analyzed a different dataset this year — there's no shared answer to group presentations around, so teams simply present in roster order. With every team across two standard 50-minute class sessions, there's no room for a debrief after each presentation. On Mon Dec 7, transitions between teams are brief (a few seconds); the full cross-team debrief happens once, at the end of Wed Dec 9, after every team has presented.

Schedule:

Day	Teams	Presentation time	Debrief
Mon Dec 7	9	45 min	Brief transitions only
Wed Dec 9	6	30 min	Full cross-team debrief (~15–18 min)

Cross-Team Summary

Built live as teams present — there's no pre-known reference table this year, since every team chose its own topic. See the cross-team findings tracker for the tracking sheet.

Come ready to compare: does a strong regression fit (high R^2) show up more in some domains than others? Whose predictor turned out to be a weaker driver than expected? Whose GROUP coefficient surprised them?

Debrief Talking Points

Used during the single full debrief at the end of Wed Dec 9, once every team has presented (not after each individual presentation — there isn't time on Mon Dec 7):

- Which team's model had the **highest R^2** ? Which had the lowest? Is that a sign of a better analysis, or just a domain where the outcome is inherently more/less predictable from the chosen predictor?
- Did anyone's **GROUP coefficient come out with a counterintuitive sign**? What was the physical/practical explanation?
- Whose **bootstrap CI was much wider than the analytical OLS CI**? What did that suggest about their data (autocorrelation, small sample, outliers)?
- Compare **Monte Carlo P10–P90 ranges across very different domains** — what does a wide range mean differently for a traffic-volume prediction vs. a streamflow prediction vs. a cooling-energy prediction?

Full-Class Debrief — Comparing Across Domains

Opening question: Every team used the exact same four-method toolkit — regression, hypothesis test, Monte Carlo, bootstrap — on completely different data. Whose relationship ($X \rightarrow Y$) was strongest? Whose was weakest? Is a weak relationship a bad result, or just an honest one?

On counterintuitive coefficients: If any team found a GROUP coefficient with an unexpected sign, invite them to walk through it live. This is usually the most interesting result of the whole session — a "wrong-looking" number that turns out to have a real physical explanation once you dig in.

Full-Class Debrief — Confounding, Wherever It Shows Up

Discussion: For teams whose GROUP variable might overlap with their predictor (X) in some systematic way (e.g., a seasonal grouping that also correlates with temperature, a time-of-day grouping that correlates with traffic), ask: did the regression coefficient differ meaningfully from the raw group mean difference? If so, that's confounding at work — the same phenomenon studied all semester, just showing up in a new dataset.

Discussion: For any team that didn't find confounding, ask how they checked for it, and whether they're confident it's genuinely absent or just not detected.

What a Shared Toolkit, Applied to Different Problems, Produced

Every team used the same formula shape: $Y \sim X + \text{GROUP}$, the same t-test, the same Monte Carlo and bootstrap methods.

The R^2 values, coefficient signs, and uncertainty ranges varied enormously — because the datasets themselves are about completely different physical and operational systems.

The math is the easy part, and it's the same math regardless of domain. The hard part is knowing which results require explanation and connecting a coefficient to a physical mechanism your client can act on — whatever that client's field turns out to be.

Course Closing

What you built this semester:

Weeks 1–4: Probability fundamentals, distributions, exceedance — in Excel

Weeks 5–8: Exploratory data analysis, visualization, midterm — dataset first contact

Weeks 9–11: Regression, hypothesis testing — statistical modeling

Weeks 12–13: Sensitivity analysis, Monte Carlo — uncertainty quantification

Week 14: Bootstrap — assumption-free inference

Week 15–16: Capstone — apply everything to a real engineering decision

Every tool you used this semester appears in professional building energy analysis, infrastructure risk assessment, and data-driven design. The methods scale from 8,760 rows to 8.76 million.